

Solution Requirements

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Input	Accept username, user ID, age, weight, fitness goal and intensity	High
2	Validation	Enforce age, weight, string length, ID pattern and intensity constraints	High
3	AI Workout Generation	Generate a personalized seven-day workout plan using Gemini	High

S.No	Requirement Category	Requirement Description	Priority
4	Nutrition / Recovery	Generate a concise tip and use a fallback when the tip service fails	High
5	Persistence	Store users and plans using SQLAlchemy and SQLite	High
6	Feedback Revision	Revise the latest plan from natural-language feedback	High
7	Original Plan Preservation	Keep the original generated plan when a revised plan exists	High
8	Saved Result	Reopen a saved plan using user ID	Medium
9	Admin View	Display saved users/ plans and support deletion	Medium
10	JSON API	Provide generation, feedback, nutrition and user retrieval endpoints	High
11	Health Endpoint	Provide /health for service status	Medium
12	Testability	Support mock-AI mode so automated tests do not require live Gemini calls	High

Step 2: Non-Functional Requirements

Category	Requirement
Usability	Clear forms, results, feedback and error messages
Reliability	AI model fallbacks and friendly service errors
Maintainability	Separate configuration, validation, database, AI and routes
Testability	Mock AI and isolated SQLite test database
Portability	Python/FastAPI application suitable for local or container deployment

Category	Requirement
Security	API keys supplied through environment variables rather than source code